



Translation and Accessibility of Multilingual Emergency Alerts

Jen Wilson

Natasha Schimka

University of Washington, Seattle

GETTING CRITICAL INFORMATION DURING THE COVID-19 PANDEMIC: EXPERIENCES OF SPANISH AND CHINESE SPEAKERS WITH LIMITED ENGLISH PROFICIENCY

Authors: [Gillian K. SteelFisher](#) , [Hannah L. Caporello](#), [Keri M. Lubell](#), [Eran N. Ben-Porath](#), [Alexander R. Green](#), [Feijun Luo](#), [Lisa Briseño](#), [Lindsay Lane](#), [Sarah E. Sheff](#), [Julio Dicient Taillepierre](#), [Laura Espino](#), and [Alyssa Boyea](#) | [AUTHORS INFO & AFFILIATIONS](#)

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Barriers and best practices for inclusive emergency alerts and warnings

Melissa Villarreal ^a  , Carson MacPherson-Krutzky ^b  , Mary Angelica Painter ^b  

¿Aviso o Alerta? Developing Effective, Inclusive, and Consistent Watch and Warning Translations for U.S. Spanish Speakers

Joseph E. Trujillo-Falcón, América R. Gaviria Pabón, Joseph T. Ripberger, Abby Bitterman, Jonathan B. Thornton, Makenzie J. Krocak, Sean R. Ernst, Estilita Cassiani Obeso, and John Lipski

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Emergency alerts go out in Spanish and English, but what happens if you speak another language?

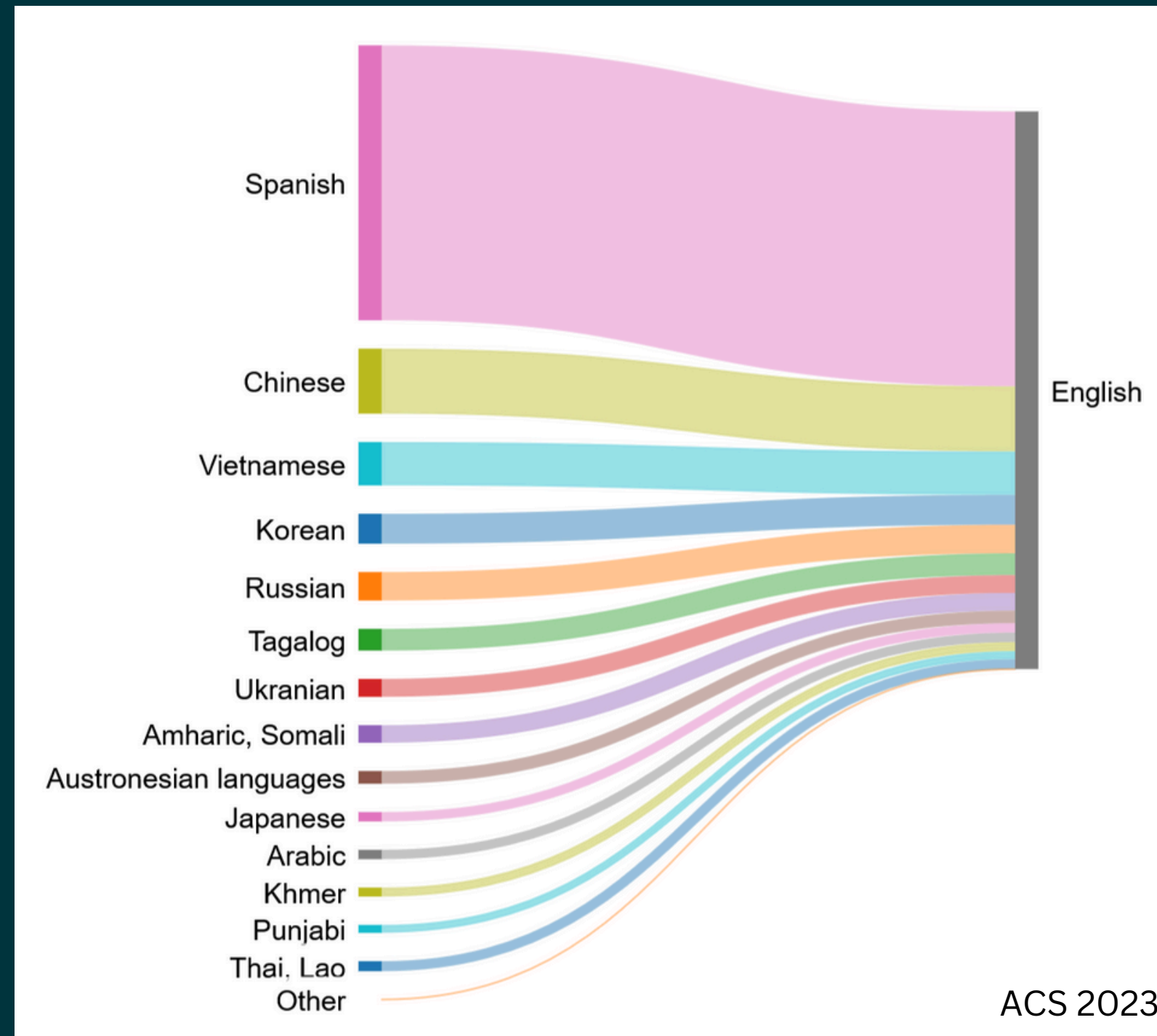
JUNE 16, 2025 · 6:00 AM ET

HEARD ON MORNING EDITION

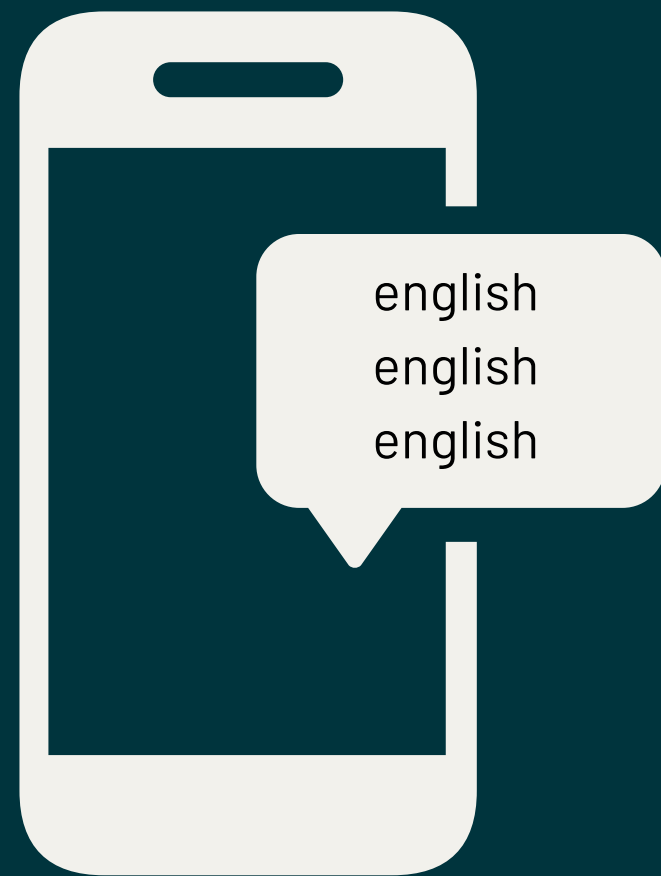
By [Nate Perez](#)

what happens when
someone
receives an alert they
cannot read?

how do we read emergency alerts?



how do we translate the messages?



do nothing

ask a trusted source

translate it

how do we get translated output?

Translation services

google translate, deepL

Generative AI (GAI) services

chatgpt, deepseek, gemini

What is GAI?

**machine learning models with probabilistic
outputs**

how do we get translated output from Generative AI?

With a service like chatgpt, deepseek, or gemini, we used prompts like these:

create this

```
Can you write an emergency text  
about DISASTER in LANGUAGE  
please?
```

```
Only return the text of the  
alert in LANGUAGE. Do not  
include any other response to  
the prompt.
```

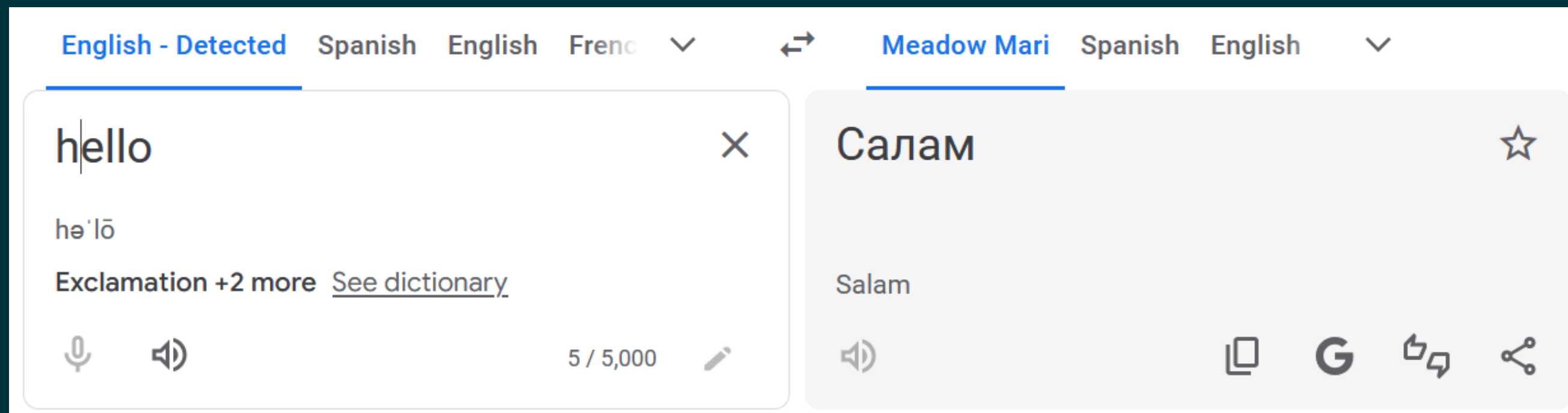
translate this

```
Translate this emergency alert  
to LANGUAGE. Don't respond  
with anything outside of the  
translation. For the variables  
in square brackets, leave those  
in English.
```

```
[SENDING AGENCY]: An EXTREME  
WIND WARNING is in effect for  
[LOCATION] for the immediate  
danger of life-threatening winds  
until [TIME]. Take cover NOW  
in an interior room of a sturdy  
building, away from windows.  
Protect your head from flying  
objects. Do NOT go outside  
if the wind calms! Winds will  
quickly become dangerous again.  
[URL]
```

how do we get translate output from translation services?

google translate, deepL



how do we evaluate translations?

gold standard

translation

how do we evaluate translations?



human
evaluation

BLEU

classic MT

ROUGE

patterns and overlap

COMET

transformer-based

BERTScore

meaning

automatically

how do we evaluate translations?

BLEU

ROUGE

classic MT
patterns and overlap

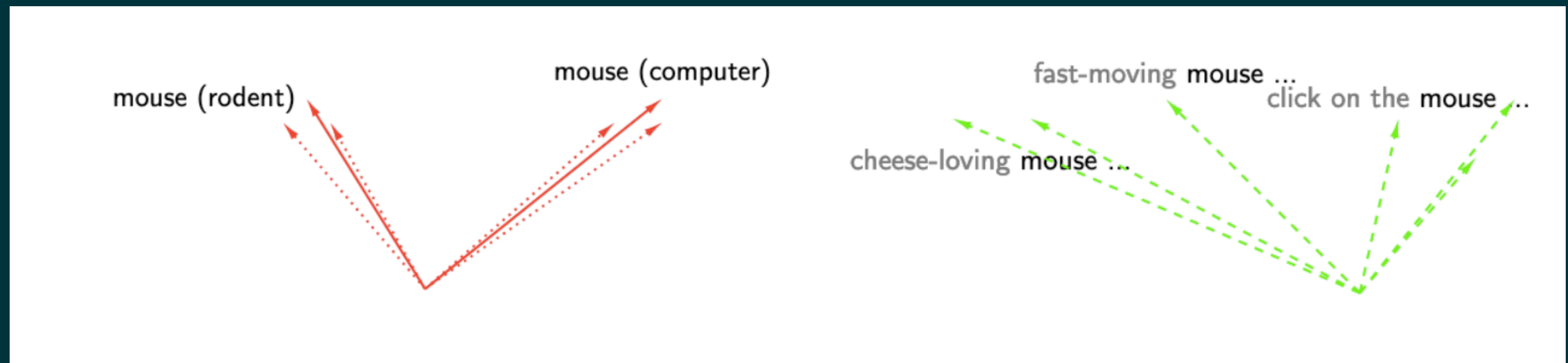
Una ADVERTENCIA DE INCENDIO en [LOCATION] está en efecto hasta [TIME]	Una ADVERTENCIA DE INCENDIO en [UBICACIÓN] está vigente hasta [HORA]
Evacúe a su familia y mascotas ahora, sin demoras.	Evacue a su familia y mascotas ahora mismo, sin demora.
La visibilidad del área disminuirá y los caminos pueden bloquearse.	La visibilidad en la zona se reducirá y las carreteras podrían quedar bloqueadas.
Si no sale ahora, podría quedar atrapado, resultar herido o morir.	Si no se marcha ahora, podría quedar atrapado, lesionado o incluso morir.
ABANDONE las áreas cercanas a [LOCATION].	ABANDONE las zonas cercanas a [UBICACIÓN].
Prepárese para una reducción de la visibilidad, denso humo y dificultad para respirar.	Prepárese para una visibilidad reducida, humo denso y dificultad para respirar.
Maneje con cuidado.	Tenga cuidado al conducir.
Preste atención al personal de seguridad pública en el área y siga sus instrucciones. [URL]	Esté atento al personal de seguridad pública que opera en la zona y siga sus instrucciones. [URL]

how do we evaluate translations?

COMET

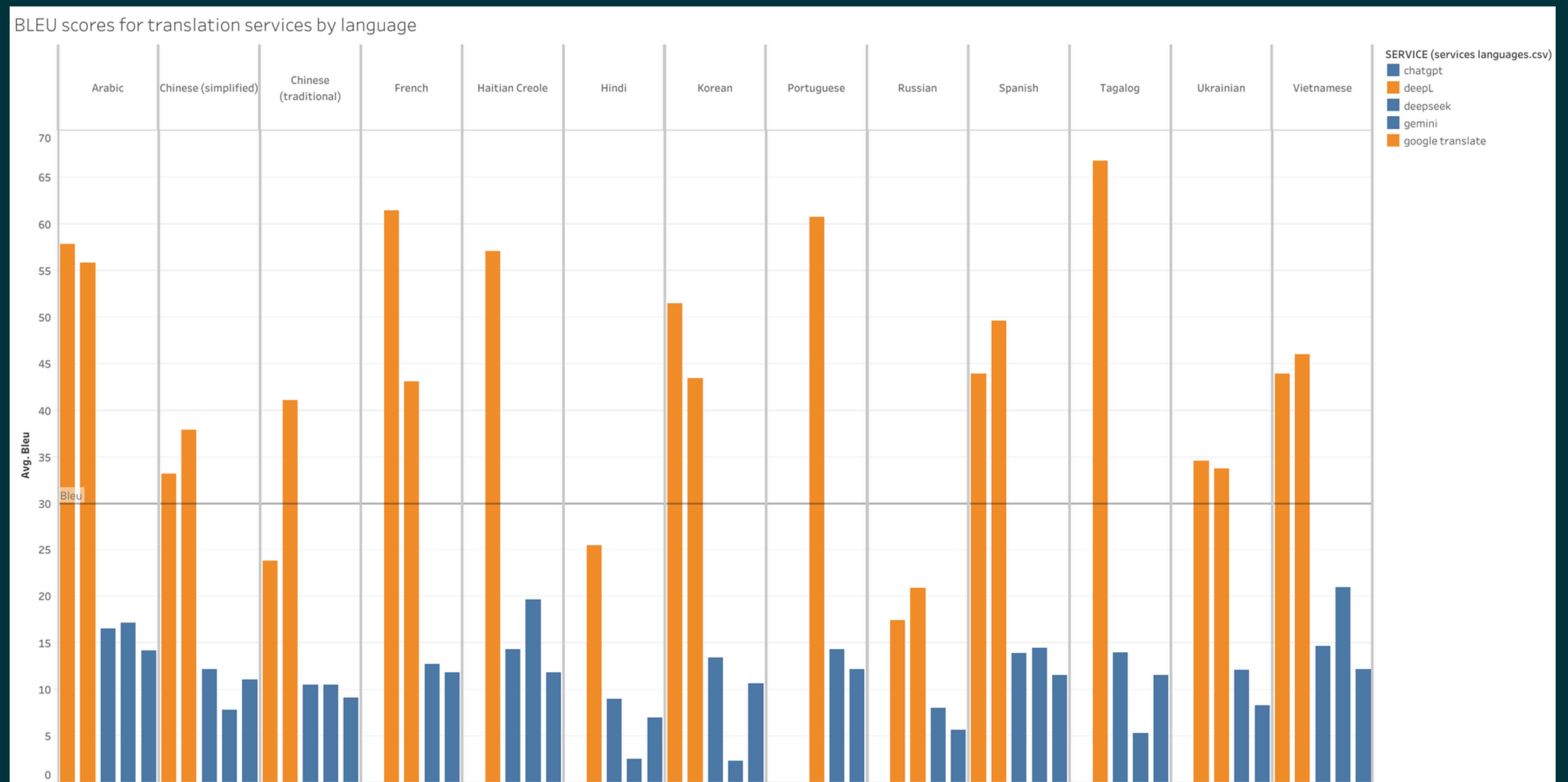
transformer-based
meaning

BERTScore



source: [Kawin Ethayarajh](#)

how do we evaluate translations?

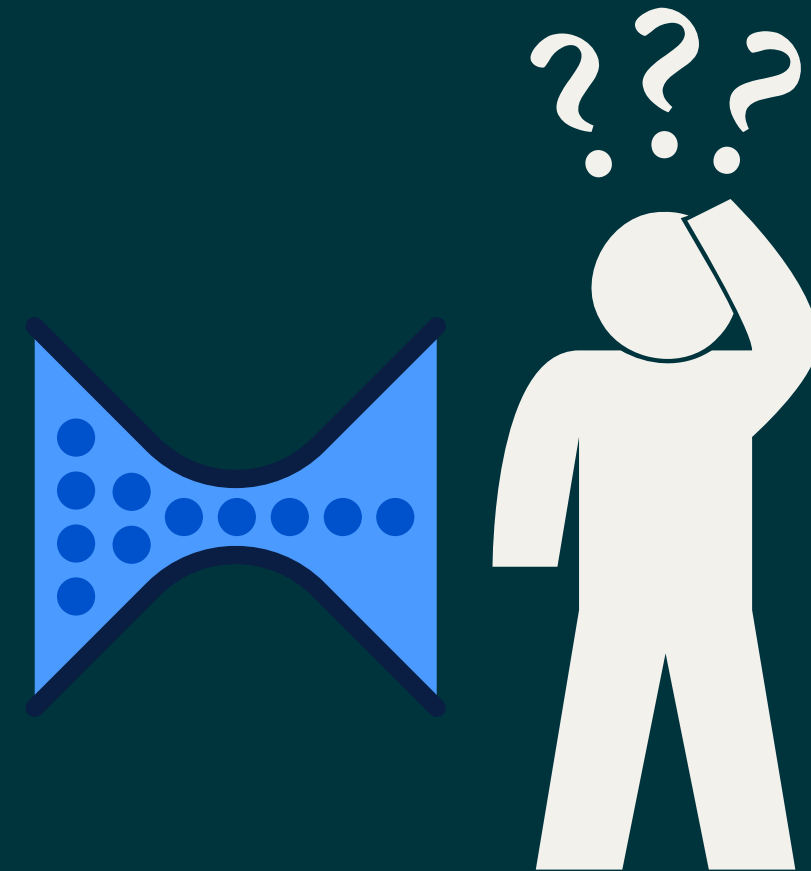
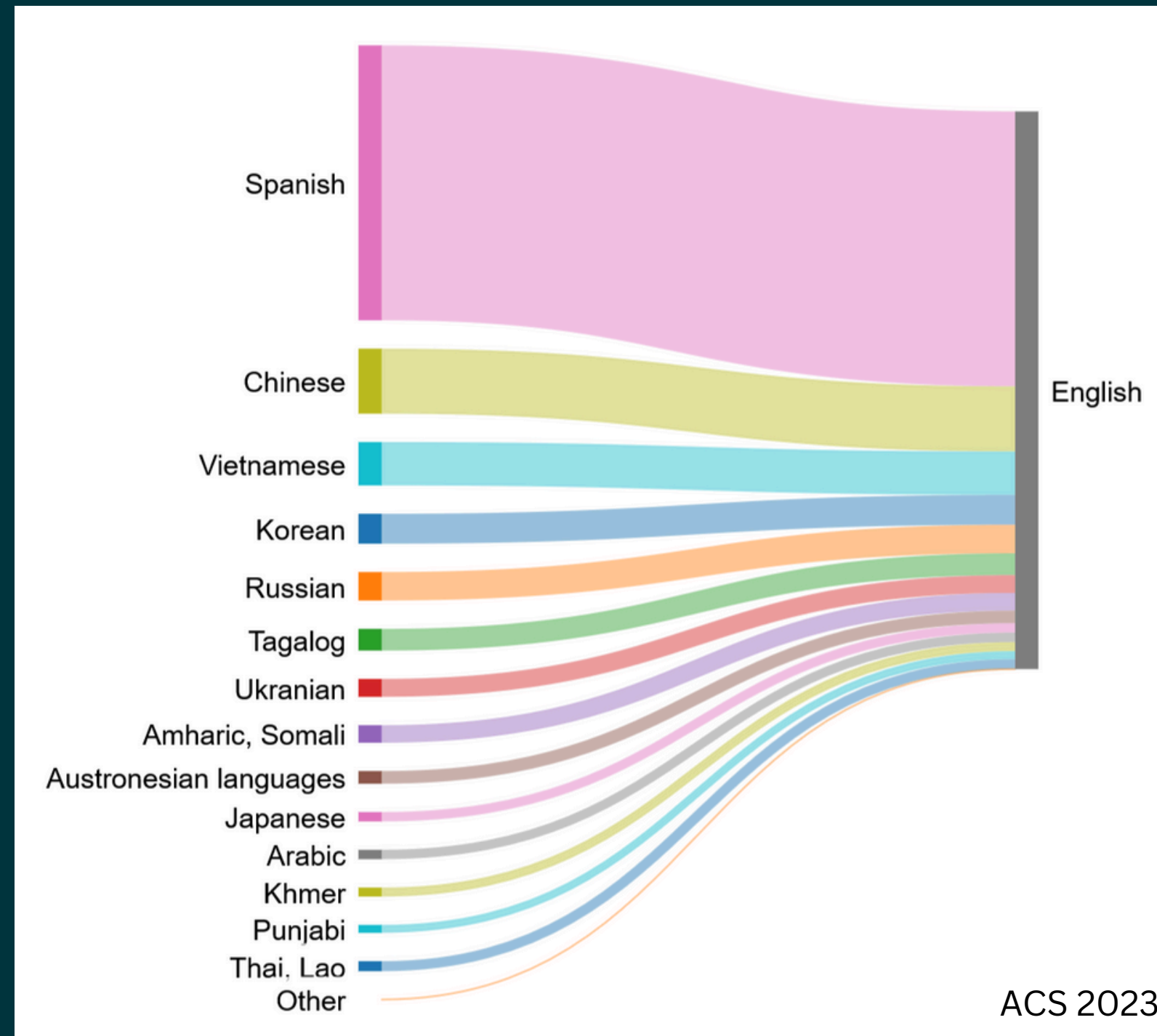


what is the best service?

It depends



how do we read emergency alerts?



IN CASE OF EMERGENCY



Individuals and families

**Use google translate because it
is good enough and available offline**

Other services if desired and available

IN CASE OF EMERGENCY



Agencies

Prepare:

- Download google translate language packs for responders
- Encourage individuals to download any language pack they may need

Provide:

- Link to pre-translated alerts

Jen Wilson

jenwils@uw.edu

